

Lecture 6 (9/25)

Variance, Normal models and Linear Regression

Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2]$$

Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2]$$

Using

$$(Y - E[Y])^2 = Y^2 - 2YE[Y] + E[Y]^2$$

Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2]$$

Using

$$(Y - E[Y])^2 = Y^2 - 2YE[Y] + E[Y]^2$$

and that expectation is a linear operation

$$\text{var}(Y) = E[Y^2] - 2E[Y]^2 + E[Y]^2 = E[Y^2] - E[Y]^2$$

Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2]$$

Using

$$(Y - E[Y])^2 = Y^2 - 2YE[Y] + E[Y]^2$$

and that expectation is a linear operation

$$\text{var}(Y) = E[Y^2] - 2E[Y]^2 + E[Y]^2 = E[Y^2] - E[Y]^2$$

In code `np.var(y)`.

Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2]$$

Using

$$(Y - E[Y])^2 = Y^2 - 2YE[Y] + E[Y]^2$$

and that expectation is a linear operation

$$\text{var}(Y) = E[Y^2] - 2E[Y]^2 + E[Y]^2 = E[Y^2] - E[Y]^2$$

In code `np.var(y)`. **Properties:**

- **Linear transformations:** For constants a and b , $\text{var}(aY + b) = a^2 \text{var}(Y)$.

Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2]$$

Using

$$(Y - E[Y])^2 = Y^2 - 2YE[Y] + E[Y]^2$$

and that expectation is a linear operation

$$\text{var}(Y) = E[Y^2] - 2E[Y]^2 + E[Y]^2 = E[Y^2] - E[Y]^2$$

In code `np.var(y)`. **Properties:**

- **Linear transformations:** For constants a and b , $\text{var}(aY + b) = a^2 \text{var}(Y)$.
- **Sums of independent variables:** If X and Y are independent, $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$.

Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2]$$

Using

$$(Y - E[Y])^2 = Y^2 - 2YE[Y] + E[Y]^2$$

and that expectation is a linear operation

$$\text{var}(Y) = E[Y^2] - 2E[Y]^2 + E[Y]^2 = E[Y^2] - E[Y]^2$$

In code `np.var(y)`. **Properties:**

- **Linear transformations:** For constants a and b , $\text{var}(aY + b) = a^2 \text{var}(Y)$.
- **Sums of independent variables:** If X and Y are independent, $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$.

Related quantities:

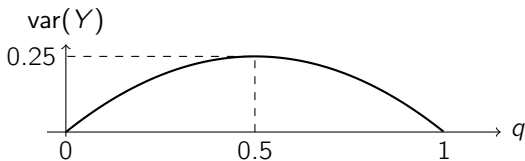
- The standard deviation is $\sigma = \sqrt{\text{var}(Y)}$
- The coefficient of variation is $\text{CV}_Y = \sigma/\mu$ where $\mu = E[Y]$.

Example

Let $Y \sim \text{Bernoulli}(q)$. Compute $\text{var}(Y)$.

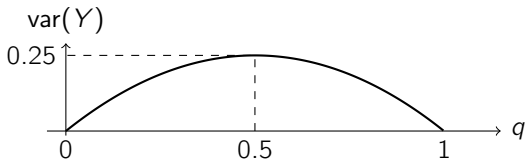
Example

Let $Y \sim \text{Bernoulli}(q)$. Compute $\text{var}(Y)$.



Example

Let $Y \sim \text{Bernoulli}(q)$. Compute $\text{var}(Y)$.



Example

Let $Y \sim \text{Uniform}(0, L)$. Compute $\text{var}(Y)$ and CV.

The Normal distribution

$Y \sim \text{Normal}(\mu, \sigma^2)$ has density

$$g(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-\mu)^2/2\sigma^2},$$

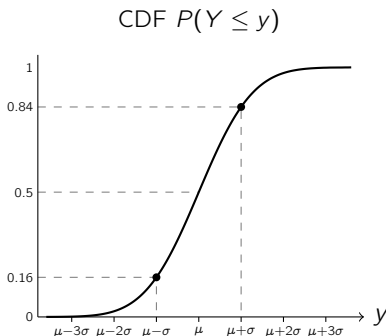
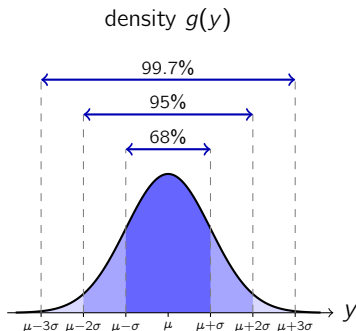
with $E[Y] = \mu$ and $\text{var}(Y) = \sigma^2$.

The Normal distribution

$Y \sim \text{Normal}(\mu, \sigma^2)$ has density

$$g(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-\mu)^2/2\sigma^2},$$

with $E[Y] = \mu$ and $\text{var}(Y) = \sigma^2$.



Properties

- The Normal distribution is the “most random”¹ random variable with a given mean and variance.

¹In the sense of max entropy

Properties

- The Normal distribution is the “most random”¹ random variable with a given mean and variance.
- If $Y \sim \text{Normal}(\mu, \sigma^2)$, then

$$Z = \frac{Y - \mu}{\sigma} \sim \text{Normal}(0, 1), \quad aY + b \sim \text{Normal}(a\mu + b, a^2\sigma^2).$$

¹In the sense of max entropy

Properties

- The Normal distribution is the “most random”¹ random variable with a given mean and variance.
- If $Y \sim \text{Normal}(\mu, \sigma^2)$, then

$$Z = \frac{Y - \mu}{\sigma} \sim \text{Normal}(0, 1), \quad aY + b \sim \text{Normal}(a\mu + b, a^2\sigma^2).$$

- If $X \sim \text{Normal}(\mu_X, \sigma_X^2)$ and $Y \sim \text{Normal}(\mu_Y, \sigma_Y^2)$ are independent, then

$$X + Y \sim \text{Normal}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$$

No other distribution with finite mean and variance has these properties!

¹In the sense of max entropy

Properties

- The Normal distribution is the “most random”¹ random variable with a given mean and variance.
- If $Y \sim \text{Normal}(\mu, \sigma^2)$, then

$$Z = \frac{Y - \mu}{\sigma} \sim \text{Normal}(0, 1), \quad aY + b \sim \text{Normal}(a\mu + b, a^2\sigma^2).$$

- If $X \sim \text{Normal}(\mu_X, \sigma_X^2)$ and $Y \sim \text{Normal}(\mu_Y, \sigma_Y^2)$ are independent, then

$$X + Y \sim \text{Normal}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$$

No other distribution with finite mean and variance has these properties!

¹In the sense of max entropy

Conditional Normal Model

A (single-predictor) linear regression model is conditionally Normal:

$$Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2).$$

Conditional Normal Model

A (single-predictor) linear regression model is conditionally Normal:

$$Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2).$$

Equivalently,

$$Y = \beta_0 + \beta_1 X + \epsilon, \quad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2).$$

Conditional Normal Model

A (single-predictor) linear regression model is conditionally Normal:

$$Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2).$$

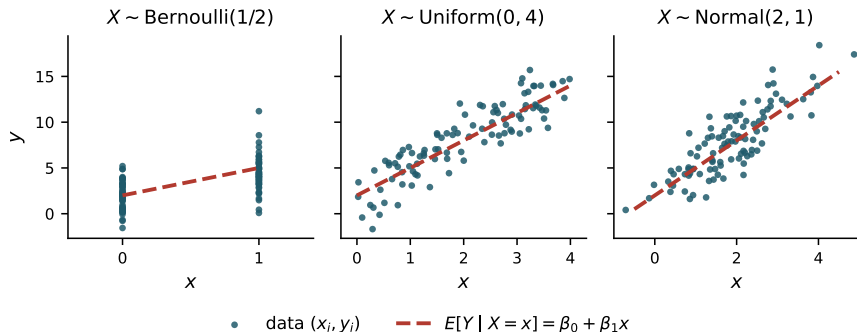
Equivalently,

$$Y = \beta_0 + \beta_1 X + \epsilon, \quad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2).$$

- X is the predictor
- Y is the response variable
- β_1 is called the regression coefficient of X .
- The variable ϵ may be referred to as an exogenous noise source. Exogeneity means this noise is independent of X .

What data from this model looks like

Data simulated from $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\varepsilon^2)$



Linear regression assumptions

In a linear regression model we make 3 assumptions:

Linear regression assumptions

In a linear regression model we make 3 assumptions:

- 1) Conditional normality: $Y | X$ is Normal.

Linear regression assumptions

In a linear regression model we make 3 assumptions:

- 1) Conditional normality: $Y \mid X$ is Normal.
- 2) Linearity of the conditional mean: $E[Y \mid X = x] = \beta_0 + \beta_1 x$.

Linear regression assumptions

In a linear regression model we make 3 assumptions:

- 1) Conditional normality: $Y | X$ is Normal.
- 2) Linearity of the conditional mean: $E[Y | X = x] = \beta_0 + \beta_1 x$. This means increasing x by one changes Y by β_1 , on average. For any x

$$\beta_1 = E[Y | X = x + 1] - E[Y | X = x]$$

Linear regression assumptions

In a linear regression model we make 3 assumptions:

- 1) Conditional normality: $Y | X$ is Normal.
- 2) Linearity of the conditional mean: $E[Y | X = x] = \beta_0 + \beta_1 x$. This means increasing x by one changes Y by β_1 , on average. For any x

$$\beta_1 = E[Y | X = x + 1] - E[Y | X = x]$$

- 3) Homoskedasticity: $\text{var}(Y | X = x)$ does not depend on x .

Marginals and covariance

Now suppose $E[X] = \mu_X$ and $\text{var}(X) = \sigma_X^2$. X can be any distribution with this mean and variance.

Marginals and covariance

Now suppose $E[X] = \mu_X$ and $\text{var}(X) = \sigma_X^2$. X can be any distribution with this mean and variance. Calculate the following:

- $\mu_Y = E[Y]$.
- $E[XY]$

Marginals and covariance

Now suppose $E[X] = \mu_X$ and $\text{var}(X) = \sigma_X^2$. X can be any distribution with this mean and variance. Calculate the following:

- $\mu_Y = E[Y]$.
- $E[XY]$

The covariance is

$$\text{cov}(X, Y) = E[XY] - E[Y]E[X]$$

Marginals and covariance

Now suppose $E[X] = \mu_X$ and $\text{var}(X) = \sigma_X^2$. X can be any distribution with this mean and variance. Calculate the following:

- $\mu_Y = E[Y]$.
- $E[XY]$

The covariance is

$$\text{cov}(X, Y) = E[XY] - E[Y]E[X]$$

and from the calculations above

$$\beta_1 = \frac{\text{cov}(X, Y)}{\sigma_X^2}$$

Marginals and covariance

Now suppose $E[X] = \mu_X$ and $\text{var}(X) = \sigma_X^2$. X can be any distribution with this mean and variance. Calculate the following:

- $\mu_Y = E[Y]$.
- $E[XY]$

The covariance is

$$\text{cov}(X, Y) = E[XY] - E[Y]E[X]$$

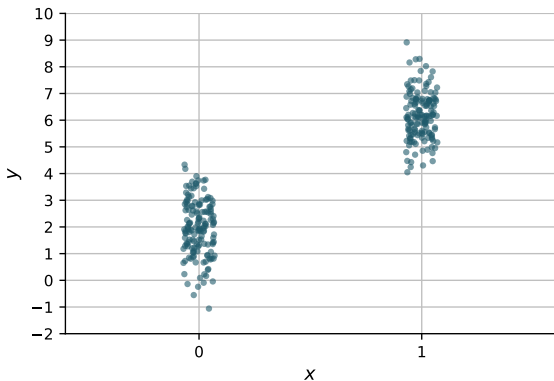
and from the calculations above

$$\boxed{\beta_1 = \frac{\text{cov}(X, Y)}{\sigma_X^2}}$$

This formula connects the regression coefficient to the covariance (and hence to expectations we can compute from data).

Example

Write down a linear regression model for the following data, where you can assume $X \sim \text{Bernoulli}(1/2)$. What is $\text{cov}(X, Y)$ and $\text{var}(Y)$?



Example

Two attributes X and Y (e.g. income and height) have a covariance of 2. An increase of X by two units leads to an increase in Y of 1, on average. Assuming a linear regression model $Y | X$, what is the variance of X ?

Marginal variance

Calculate $\sigma_Y^2 = \text{var}(Y)$, where $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$

$$\text{var}(Y) = \text{var}(\beta_0 + \beta_1 X + \epsilon)$$

Marginal variance

Calculate $\sigma_Y^2 = \text{var}(Y)$, where $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$

$$\begin{aligned}\text{var}(Y) &= \text{var}(\beta_0 + \beta_1 X + \epsilon) \\ &= \text{var}(\beta_1 X + \epsilon) \quad (\text{adding a constant doesn't change the variance})\end{aligned}$$

Marginal variance

Calculate $\sigma_Y^2 = \text{var}(Y)$, where $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$

$$\begin{aligned}\text{var}(Y) &= \text{var}(\beta_0 + \beta_1 X + \epsilon) \\ &= \text{var}(\beta_1 X + \epsilon) && \text{(adding a constant doesn't change the variance)} \\ &= \text{var}(\beta_1 X) + \text{var}(\epsilon) && \text{(variances add for independent variables)}\end{aligned}$$

Marginal variance

Calculate $\sigma_Y^2 = \text{var}(Y)$, where $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$

$$\begin{aligned}\text{var}(Y) &= \text{var}(\beta_0 + \beta_1 X + \epsilon) \\ &= \text{var}(\beta_1 X + \epsilon) && \text{(adding a constant doesn't change the variance)} \\ &= \text{var}(\beta_1 X) + \text{var}(\epsilon) && \text{(variances add for independent variables)} \\ &= \beta_1^2 \text{var}(X) + \text{var}(\epsilon) && (\text{var}(aX) = a^2)\end{aligned}$$

Marginal variance

Calculate $\sigma_Y^2 = \text{var}(Y)$, where $Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$

$$\begin{aligned}\text{var}(Y) &= \text{var}(\beta_0 + \beta_1 X + \epsilon) \\&= \text{var}(\beta_1 X + \epsilon) && \text{(adding a constant doesn't change the variance)} \\&= \text{var}(\beta_1 X) + \text{var}(\epsilon) && \text{(variances add for independent variables)} \\&= \beta_1^2 \text{var}(X) + \text{var}(\epsilon) && (\text{var}(aX) = a^2) \\&= \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2 && \text{(definitions of } \sigma_X^2 \text{ and } \sigma_\epsilon^2)\end{aligned}$$

Standardized variables

Let $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$ be two random variables related by a linear regression model.

Standardized variables

Let $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$ be two random variables related by a linear regression model.

Consider the standardized variables

$$Z_X = (X - \mu_X)/\sigma_X, \quad Z_Y = (Y - \mu_Y)/\sigma_Y$$

What is the regression coefficient of Z_X on Z_Y ?

Standardized variables

$$Z_Y = (Y - \mu_Y) / \sigma_Y$$

Standardized variables

$$\begin{aligned} Z_Y &= (Y - \mu_Y) / \sigma_Y \\ &= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)] / \sigma_Y \quad (\text{substitute } Y, \mu_Y) \end{aligned}$$

Standardized variables

$$\begin{aligned}Z_Y &= (Y - \mu_Y)/\sigma_Y \\&= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)]/\sigma_Y \quad (\text{substitute } Y, \mu_Y) \\&= [\cancel{\beta_0} + \beta_1 X + \epsilon - \cancel{\beta_0} - \beta_1 \mu_X]/\sigma_Y \quad (\beta_0 \text{ cancels})\end{aligned}$$

Standardized variables

$$\begin{aligned}Z_Y &= (Y - \mu_Y)/\sigma_Y \\&= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)]/\sigma_Y && \text{(substitute } Y, \mu_Y) \\&= [\cancel{\beta_0} + \beta_1 X + \epsilon - \cancel{\beta_0} - \beta_1 \mu_X]/\sigma_Y && (\beta_0 \text{ cancels}) \\&= [\beta_1(X - \mu_X) + \epsilon]/\sigma_Y && \text{(factor out } \beta_1)\end{aligned}$$

Standardized variables

$$\begin{aligned}Z_Y &= (Y - \mu_Y)/\sigma_Y \\&= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)]/\sigma_Y && \text{(substitute } Y, \mu_Y) \\&= [\cancel{\beta_0} + \beta_1 X + \epsilon - \cancel{\beta_0} - \beta_1 \mu_X]/\sigma_Y && (\beta_0 \text{ cancels}) \\&= [\beta_1(X - \mu_X) + \epsilon]/\sigma_Y && \text{(factor out } \beta_1) \\&= \frac{\beta_1 \sigma_X}{\sigma_Y} \cdot \frac{X - \mu_X}{\sigma_X} + \frac{\epsilon}{\sigma_Y} && \text{(multiply, divide by } \sigma_X)\end{aligned}$$

Standardized variables

$$\begin{aligned}Z_Y &= (Y - \mu_Y)/\sigma_Y \\&= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)]/\sigma_Y && \text{(substitute } Y, \mu_Y) \\&= [\cancel{\beta_0} + \beta_1 X + \epsilon - \cancel{\beta_0} - \beta_1 \mu_X]/\sigma_Y && (\beta_0 \text{ cancels}) \\&= [\beta_1(X - \mu_X) + \epsilon]/\sigma_Y && \text{(factor out } \beta_1) \\&= \frac{\beta_1 \sigma_X}{\sigma_Y} \cdot \frac{X - \mu_X}{\sigma_X} + \frac{\epsilon}{\sigma_Y} && \text{(multiply, divide by } \sigma_X) \\&= \frac{\beta_1 \sigma_X}{\sigma_Y} Z_X + \frac{\epsilon}{\sigma_Y} && (Z_X = (X - \mu_X)/\sigma_X)\end{aligned}$$

Standardized variables

$$\begin{aligned}Z_Y &= (Y - \mu_Y)/\sigma_Y \\&= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)]/\sigma_Y && \text{(substitute } Y, \mu_Y) \\&= [\cancel{\beta_0} + \beta_1 X + \epsilon - \cancel{\beta_0} - \beta_1 \mu_X]/\sigma_Y && (\beta_0 \text{ cancels}) \\&= [\beta_1(X - \mu_X) + \epsilon]/\sigma_Y && \text{(factor out } \beta_1) \\&= \frac{\beta_1 \sigma_X}{\sigma_Y} \cdot \frac{X - \mu_X}{\sigma_X} + \frac{\epsilon}{\sigma_Y} && \text{(multiply, divide by } \sigma_X) \\&= \frac{\beta_1 \sigma_X}{\sigma_Y} Z_X + \frac{\epsilon}{\sigma_Y} && (Z_X = (X - \mu_X)/\sigma_X)\end{aligned}$$

Therefore

$$\boxed{\beta_1 \sigma_X / \sigma_Y = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}}$$

This is called the correlation coefficient and denoted ρ .

Solutions

- 1) $Y \sim \text{Bernoulli}(q)$: $\text{var}(Y) = q(1 - q)$.
- 2) $Y \sim \text{Uniform}(0, L)$: $\text{var}(Y) = \frac{L^2}{12}$, $\text{CV} = \frac{L/\sqrt{12}}{L/2} = \frac{1}{\sqrt{3}} \approx 0.58$.
- 3) $\mu_Y = \beta_0 + \beta_1 \mu_X$ and $E[XY] = \beta_0 \mu_X + \beta_1(\sigma_X^2 + \mu_X^2)$, so $\text{cov}(X, Y) = \beta_1 \sigma_X^2$.
- 4) The group means are about ≈ 2 and 6 , so $\beta_1 \approx 4$. $\sigma_X^2 = \frac{1}{4}$ and $\sigma_\epsilon \approx 1$. Hence $\text{cov}(X, Y) \approx \beta_1 \sigma_X^2 = 1$ and $\text{var}(Y) \approx \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2 = 5$.
- 5) $\beta_1 = \frac{1}{2}$, so $\sigma_X^2 = \text{cov}(X, Y)/\beta_1 = 4$.